# $\ell_\infty$ norm SVP in $\widetilde{O}(3^n)$

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**Abstract**

We give a classical randomized algorithm for the Shortest Vector Problem in the infinity norm in $\widetilde{O}(3^n)$ time and space, improving the previous bound of $2^{2.443n+o(n)}$.

We use the counting moment framework of Gaussian difference dynamics to construct lists with equal expected counts at each lattice point in successively smaller cubes. The key is to restrict the midpoint of each pair to a fixed inner cube, so that every allowed difference has the same region of possible midpoints. Averaging random lattice translates then preserves these expected counts through repeated differencing. We control the dependence between entries through bounds on the second counting moment, using a weighted packing estimate for cubes to bound the variation in midpoint counts. Finally, two searches based on coordinate comparisons generate the intermediate lists and recover a shortest lattice vector within the claimed time bound.

## 1 Introduction

Let $L = B\mathbb{Z}^n \subset \mathbb{R}^n$ be a full-rank lattice. The Shortest Vector Problem in the infinity norm asks for a nonzero vector $v \in L$ satisfying

$$\|v\|_\infty = \lambda_1^{(\infty)}(L) := \min_{x \in L\setminus\{0\}} \max_{1\le i\le n} |x_i|.$$

Equivalently, it asks for the smallest cube centered at the origin that contains a nonzero lattice point. The infinity norm naturally describes coordinatewise error bounds in cryptanalytic lattice problems, including the Hidden Number Problem [BV96] and attacks on DSA and ECDSA with partially known nonces [HGS01, NS03].

Provable sieving algorithms for this norm exploit the product geometry of the cube. Aggarwal and Mukhopadhyay [AM18] specialized the AKS framework to this geometry and obtained a randomized algorithm with $2^{2.8200n+o(n)}$ time and space. Mukhopadhyay [Muk21] subsequently improved both bounds to $2^{2.4430n+o(n)}$ through a finer subdivision of the cube.

In the Euclidean setting, Gao, Han, and Hu [GHH26] introduced Gaussian difference dynamics (GDD), a counting framework for repeated differencing. It yields classical randomized algorithms for discrete Gaussian sampling at widths a fixed factor above the smoothing parameter in $(3/2)^{n/2+o(n)}$ time, with exponent approximately 0.2925, and for exact Euclidean SVP in $2^{0.4130n+o(n)}$ time, both on arbitrary lattices. The core idea is to maintain bounds on the first two counting moments throughout the list iteration. The first describes the expected number of entries at each lattice point relative to a Gaussian reference, while the second counts ordered pairs of distinct entries and limits their clustering. Conditional independence between separately generated child lists gives a

closed recursion for these moments even though entries within either list may share inputs and ancestors. The resulting bounds track the supply of useful vectors through successive rounds and connect expected target counts to the probability of success.

The moment recursion itself is general, but the efficient Euclidean constructions rely on Gaussian completion of squares, thin spherical shells, and geometric searches for short differences. These ingredients both preserve the Gaussian reference through the iteration and determine the cost of generating the next list. For the infinity norm, we instead seek equal expected counts at the lattice points of each successive cube. Directly restricting a difference to a smaller cube does not preserve this reference: its region of possible midpoints varies with the difference vector and shrinks toward the boundary. Controlling the first two counting moments therefore calls for a cube transition with suitable representation counts and variance bounds, together with searches adapted to that transition.

We give a classical randomized algorithm for exact $\ell_\infty$-SVP in $\widetilde{O}(3^n)$ time and space, reducing the exponent from 2.4430 to $\log_2 3 \approx 1.5850$. Restricting every retained pair to the same midpoint cube makes the midpoint volume independent of the allowed difference, and averaging a random lattice translate gives exact first-moment closure. A weighted packing bound with factor $(9/8)^n$, combined with a random superlattice, controls the variation in midpoint counts and determines the required list size. Finally, ternary coordinate classes generate the intermediate lists, while a binary condition combined with quotient labels recovers a shortest vector. The candidate and report bounds put both searches within $\widetilde{O}(3^n)$ time. Throughout the paper, $\widetilde{O}$ suppresses factors polynomial in the dimension and the binary input length.

### 1.1 Our results

**Theorem 1.1** (Main theorem)**.** *There is a classical randomized algorithm that, given a nonsingular rational basis of a full-rank lattice $L \subset \mathbb{R}^n$, always returns a nonzero vector of $L$, which is shortest in the infinity norm with probability at least $2/3$. Every execution uses*

$$T = \widetilde{O}(3^n) \qquad \text{and} \qquad S = \widetilde{O}(3^n).$$

*The success probability can be increased to $1 - 2^{-\Omega(n \log n)}$ by independent repetition without changing either bound.*

**Application to Subset Balancing.** Combining Theorem 1.1 with the reduction of Gao, Feng, Hu, and Pan [GFHP26, Theorem 3.3] to one $(n+1)$-dimensional $\ell_\infty$-SVP instance gives the following consequence.

**Corollary 1.2.** *Given $x \in \mathbb{Z}^n$ and an integer $d \geq 1$, there is a randomized algorithm that, with probability at least $2/3$, finds a vector $c \in \{-d, \ldots, d\}^n \setminus \{0\}$ satisfying $c \cdot x = 0$, or correctly reports that no such vector exists. It uses $\widetilde{O}(3^n)$ time and space.*

For every $d \geq 5$, this is exponentially faster than the standard meet-in-the-middle bound $\widetilde{O}((2d+1)^{n/2})$, improving the threshold $d \geq 15$ obtained in [GFHP26].

For every fixed $d \geq 5$, our exponent is smaller than the optimized bound from the analysis of Randolph and Węgrzycki [RW26], which at $d = 5$ is approximately 1.60872, compared with our $\log_2 3 \approx 1.58496$. As far as we know, we thus obtain the best known worst case running time among classical algorithms for Subset Balancing in this range.

### 1.2 Related work

Ajtai, Kumar, and Sivakumar introduced randomized sieving for lattice problems [AKS01]. Algorithms in general norms were subsequently developed through sieving and through enumeration. Dadush, Peikert, and Vempala obtained single-exponential algorithms for SVP in arbitrary norms by using M-ellipsoid coverings [DPV11].

The infinity norm bounds discussed above use linear sieves adapted to the cube. Mukhopadhyay's sieve [Muk21] also applies to every $\ell_p$ norm, with a general SVP bound of $2^{2.7510n+o(n)}$ time and space, and gives approximation algorithms for SVP and CVP in these norms.

Random superlattices and quotient labels appear in the Euclidean algorithms of Gao, Feng, and Hu [GFH26a, GFH26b]. A superlattice contains every target vector of the input lattice, and equal labels certify that a difference lies in the input lattice.

**Organization.** Section 2 explains the construction and derives the main cost balance. Section 3 fixes notation and the standard lattice tools. Sections 4 and 5 control cube counts on a random superlattice. Section 6 builds the lists and proves their counting bounds. Sections 7 and 8 give the two pair searches. Section 9 assembles the algorithm and proves Theorem 1.1. Section 10 discusses the connection with lattice kissing numbers. Appendix A proves the basis preprocessing lemma, and Appendix B gives the finite arithmetic details.

The author used generative AI to assist with algorithm design, mathematical analysis, and manuscript drafting, as described in the AI methodology statement at the end of the paper.

## 2 Technique overview

Fix a shortest vector $v \in L \setminus \{0\}$ for the analysis, and write $\lambda = \|v\|_\infty$. There are two tasks: make $v$ appear as a difference between two lists, and find that difference within the claimed time. We first explain the construction assuming that $\lambda$ is known, then remove this assumption by trying polynomially many scales.

The lists are generated using the counting recursion of GDD [GHH26]. At each level, we preserve equal expected counts at the allowed lattice points and bound the expected number of ordered pairs of distinct entries. These two bounds track the available supply and control clustering, allowing us to turn a large expected number of target representations into a high success probability.

### 2.1 Preserving uniform counts by midpoint restriction

We seek a difference rule that preserves equal expected counts across each successive cube. Write

$$Q_R = [-R, R)^n, \qquad B_R = [-R/2, R/2)^n.$$

The input lists lie in $Q_R$, while $B_R$ is the allowed midpoint region. For a fixed $\gamma < 1$, keep pairs from the two lists satisfying

$$z = x - y \in Q_{\gamma R}, \qquad M = (x + y)/2 \in B_R. \tag{1}$$

For every allowed $z$, the complete set of possible midpoints is exactly $B_R$. Conversely, $M \in B_R$ and $z \in Q_{\gamma R}$ imply $M \pm z/2 \in Q_R$. The volume of the midpoint set is therefore $R^n$, independently of $z$.

Restricting only $x - y$ would give a midpoint region that shrinks as a coordinate of $z$ approaches the boundary. The midpoint condition in (1) removes this variation, giving a constant output density on $Q_{\gamma R}$ in the continuous model.

For lattice points, the number of admissible midpoints also depends on the lattice translate. We build the lists along a binary tree, choosing a new random translate at each node and generating the two child lists with independent randomness conditional on their translates. Averaging the translate makes the expected output count equal at every lattice point of the smaller cube. The second counting moment accounts for both the variation in midpoint counts and the dependence caused by pairs that share an input entry. Its recursion requires a bound on the midpoint count's variance, which we obtain by passing to a random superlattice.

### 2.2 Controlling midpoint variance with a random superlattice

A cube near the shortest vector scale can contain very different numbers of points in different translates of the input lattice. For a lattice $\Lambda$ and a uniform translate $t + \Lambda$, let

$$J_R^\Lambda(t) = \#((t + \Lambda) \cap B_R).$$

We measure this variation by

$$\Theta_R(\Lambda^*) = \frac{\mathrm{Var}(J_R^\Lambda)}{\mathbb{E}[J_R^\Lambda]^2}, \qquad \mathbb{E}[J_R^\Lambda] = \frac{R^n}{\det \Lambda}. \tag{2}$$

We work in a random superlattice $\Gamma \supset L$, of determinant $D$, chosen so that these relative variances are small at all widths used by the construction.

The main geometric estimate is

$$1 + \Theta_R(L^*) \le C^n \frac{\det L}{\lambda^n}, \qquad C = \frac{9}{8}, \qquad R \ge \frac{8}{9}\lambda. \tag{3}$$

Its proof is a weighted packing argument for cubes. The constant can already be seen in one dimension. An interval of length $4/3$, translated uniformly modulo $\mathbb{Z}$, contains one point with probability $2/3$ and two points with probability $1/3$. Its second moment divided by the square of its mean is $9/8$. Products of these intervals give $(9/8)^n$.

Passing to a random superlattice of prime index $p$ reduces the expected relative variance to at most $\Theta_R(L^*)/p$, plus an error term controlled in Section 5. We choose

$$p \asymp C^n \frac{\det L}{\lambda^n} \operatorname{poly}(n), \qquad D \asymp \lambda^n C^{-n} / \operatorname{poly}(n).$$

This makes the relative variance small at all required widths with high probability. The density also determines the list size: for lists of a fixed average size, decreasing $D$ decreases the expected number of representations of the target vector, as the following calculation shows.

### 2.3 Choosing the list size from target representations

Suppose a final list has expected count $A$ at each point of a common translate $z + \Gamma$ inside $Q_\lambda$. If its average size is $N$, then

$$A = \frac{ND}{(2\lambda)^n}.$$

Every midpoint in $B_\lambda$ represents $v$ through the endpoints $M + v/2$ and $M - v/2$. The number of available midpoints is about $\lambda^n/D$. Thus the expected number of representations in the two lists is

$$A^2 \frac{\lambda^n}{D} = \frac{N^2 D}{4^n \lambda^n} \asymp \frac{N^2}{4^n C^n \operatorname{poly}(n)}. \tag{4}$$

This suggests

$$N = (2\sqrt{C})^n \operatorname{poly}(n) = (3/\sqrt{2})^n \operatorname{poly}(n).$$

A suitable polynomial factor makes the target count large. To turn this expectation into a success probability, we also control pairs of distinct entries within each list. Repeated vector values are kept as separate entries, so that every retained pair contributes one new entry to the counting recursion.

At every round, the reference count per point is $A_R = ND/(2R)^n$. We thin each eligible pair with probability

$$q = \frac{(2/\gamma)^n}{N}.$$

Then $qA_R^2(R^n/D) = A_{\gamma R}$, so the same reference list size $N$ is maintained throughout the tree. We choose a fixed $\sqrt{8/9} < \gamma < 1$, which ensures $q \le 1$ while shortening the radius by a constant factor.

The construction starts in a cube whose radius is polynomially larger than $\lambda$, where we can sample the initial lists directly with the required counting moments. The tree therefore has depth $O(\log n)$ and polynomially many nodes, so its leading exponential cost is determined by the list size and the pair searches at each node.

### 2.4 Finding pairs with coordinatewise tests

Gaussian difference dynamics [GHH26] uses BDGL product codes [BDGL16] to find pairs with short Euclidean differences. In the infinity norm, the difference and midpoint conditions can be tested coordinate by coordinate. We exploit this structure directly: three classes per coordinate generate candidates for each recursive merge, while a binary condition and quotient labels recover a shortest vector from the two root lists.

For an internal pair, (1) implies

$$|x_i| + |y_i| = \max\{|x_i - y_i|, |x_i + y_i|\} \le R.$$

Divide each absolute coordinate into three classes, according to the intervals $[0, R/3]$, $(R/3, 2R/3]$, and $(2R/3, R]$. If the class numbers are $b_i(x), b_i(y) \in \{0, 1, 2\}$, every eligible pair satisfies

$$b_i(x) + b_i(y) \le 2.$$

There are six compatible class pairs out of nine in each coordinate. A table indexed by $\{0, 1, 2\}^n$ reports the corresponding pairs without visiting empty combinations. The candidate count has scale

$$N^2(2/3)^n = 3^n \operatorname{poly}(n).$$

These tables also account for the $\widetilde{O}(3^n)$ space bound, which exceeds the space needed to store the lists.

At the root, the task is to recover a vector of $L$ from the two lists. Equal quotient labels recognize differences in $L$. We combine this condition with a binary rule: in every coordinate, at least one endpoint must lie in $[-\lambda/2, \lambda/2)$. This keeps every representation of a shortest vector whose midpoint lies in $B_\lambda$, since $|v_i| \le \lambda$ ensures that at least one of $M_i + v_i/2$ and $M_i - v_i/2$ lies in the inner interval.

The binary rule also lets us use the separation of $L$ to bound the number of reports. In one dimension, inclusion and exclusion rewrites the compatibility count as the overlap of the inner interval with a nonnegative weight. This weight is one on the inner interval and two on the remainder of $[-\lambda, \lambda)$, so its total mass is $\lambda + 2\lambda = 3\lambda$. Taking products over coordinates gives a weighted count

in translates of $B_\lambda$. Each such translate contains at most one point of $L$, which bounds the lattice sum by the total weight $(3\lambda)^n$. Thus, if $P$ counts pairs satisfying both conditions, then

$$\mathbb{E}[P] \le \frac{A^2(3\lambda)^n}{D} = 3^n \operatorname{poly}(n).$$

To find these pairs, we check small label groups directly and use a binary prefix table for large groups. Splitting at group size $2^{n/2}$ gives the deterministic bound

$$\operatorname{poly}(n)(M2^{n/2} + P),$$

where $M$ is the total input size. Since $M$ is at most $(3/\sqrt{2})^n \operatorname{poly}(n)$, this cost is also $3^n \operatorname{poly}(n)$.

| Part of the algorithm | Quantity before substitution | Exponential factor |
|---|---|---|
| List generation | $N$ | $(3/\sqrt{2})^n$ |
| Internal candidate pairs | $N^2(2/3)^n$ | $3^n$ |
| Final searches within label groups | $N2^{n/2}$ | $3^n$ |
| Final pair reports | $A^2(3\lambda)^n/D$ | $3^n$ |

Table 1: The cost balance. Polynomial factors are omitted in this overview.

# 3 Preliminaries

We use log for the natural logarithm and $\log_2$ for the binary logarithm. For a real number $t$, write $(t)_+ = \max\{t, 0\}$. All lists are finite sequences of entries with distinct identifiers. Different entries may store the same vector.

## 3.1 Lattices and basic tools

For a nonsingular rational matrix $B$, let

$$L = B\mathbb{Z}^n, \qquad \det L = |\det B|, \qquad L^* = B^{-T}\mathbb{Z}^n.$$

We write $\lambda_i^{(p)}(L)$ for the $i$th successive minimum in the $\ell_p$ norm. Unless a different lattice is specified, $\lambda$ denotes $\lambda_1^{(\infty)}(L)$. The inequalities

$$\|x\|_\infty \le \|x\|_2 \le \sqrt{n}\|x\|_\infty$$

will be used for preprocessing only.

Minkowski's second theorem, applied to the cube, gives [Henk02]

$$\prod_{i=1}^{n} \lambda_i^{(\infty)}(L) \le \det L, \qquad \lambda_n^{(\infty)}(L) \le \frac{\det L}{\lambda^{n-1}}. \tag{5}$$

In particular, $\det L \ge \lambda^n$. We also use the Euclidean transference bound [Ban93]

$$\lambda_n^{(2)}(\Lambda)\lambda_1^{(2)}(\Lambda^*) \le n. \tag{6}$$

The following basis lemma places the starting cube within a polynomial factor of the final scale. We use the general Euclidean preprocessing result of [GHH26, Proposition 3.2].

**Lemma 3.1.** *Put $A_{\mathrm{red}}(n) = 2(n+1)$. From a rational basis of a rank-$n$ lattice $\Lambda$, one can compute a size-reduced basis $(b_1, \dots, b_n)$ such that*

$$\max_i \|b_i^*\|_2 \le A_{\mathrm{red}}(n)\lambda_n^{(2)}(\Lambda).$$

*The time and workspace are $2^{n+o(n)}\,\mathrm{poly}(\mathrm{size}(B))$. All projected bases, lifted vectors, and integer coordinates have polynomial encoding length.*

*Proof.* Taking $k = \lfloor n/2 \rfloor$ in [GHH26, Proposition 3.2] gives the claim for $n \ge 4$, and the remaining dimensions use exact SVP. Appendix A gives the proof from [ALNS20, MV13]. □

Here $b_i^*$ are the Gram–Schmidt vectors. For the computed basis matrix $B_{\mathrm{red}} = (b_1, \dots, b_n)$, size reduction gives

$$\sup_{x \in B_{\mathrm{red}}[0,1)^n} \|x\|_\infty \le \sum_i \|b_i\|_2 \le n^2 A_{\mathrm{red}}(n)\lambda_n^{(2)}(\Lambda). \tag{7}$$

The coarse factor $n^2$ is convenient and sufficient.

### 3.2 Boxes and finite shifts

We use the half-open boxes

$$Q_R = [-R, R)^n, \qquad B_R = [-R/2, R/2)^n.$$

Thus $Q_R$ has radius $R$ and $B_R$ has side length $R$. Half-open endpoints assign every grid point to exactly one interval when we partition a coordinate.

All random shifts are chosen from a finite quotient. For a fixed rational superlattice $\Gamma$, choose a rational $\delta > 0$ such that

$$\Gamma \subseteq G := \delta\mathbb{Z}^n. \tag{8}$$

Then $G/\Gamma$ has $D/\delta^n$ elements, where $D = \det\Gamma$. A uniform shift $a \in G/\Gamma$ satisfies

$$\mathbb{P}[x \in a + \Gamma] = \frac{\delta^n}{D} \qquad (x \in G). \tag{9}$$

A basis in Hermite normal form permits uniform quotient sampling in polynomial time. We describe the exact arithmetic in Appendix B.

### 3.3 Counts in random lists

For a list $\mathcal{X}$, let $N_x$ be the number of entries equal to $x$. Define

$$m_{\mathcal{X}}(x) = \mathbb{E}[N_x], \qquad S_{\mathcal{X}}(x, x') = \mathbb{E}\big[N_x N_{x'} - \mathbf{1}_{x=x'} N_x\big]. \tag{10}$$

The second quantity counts ordered pairs of distinct entries. In particular, $S_{\mathcal{X}}(x, x) = \mathbb{E}[N_x(N_x - 1)]$.

At a node with radius $R$ and prescribed coset $c + \Gamma$, the reference is

$$A_R = \frac{ND}{(2R)^n}, \qquad m_{\mathcal{X}}(x) = A_R, \qquad S_{\mathcal{X}}(x, x') \le e^{\varepsilon} A_R^2 \quad (x, x' \in (c+\Gamma) \cap Q_R). \tag{11}$$

The moments in (11) average the node's private randomness and keep its coset fixed. The first equality gives the expected supply at each point, while the second inequality limits clustering. For example, $N$ independent draws and $N$ copies of one draw have the same first moment, but very different second moments.

When a node chooses the two child cosets, we first fix those cosets and multiply the moments of the independent child subtrees, then average the common choice of child cosets. This order will be used in both the list analysis and the work estimates.

# 4 Weighted cube counts

The number of points in a translated cube is the geometric quantity needed by the list construction. We first express its variance as a nonnegative Fourier sum, then bound that sum by a packing argument in the original lattice.

## 4.1 The variance of a translated box

For a full-rank lattice $\Lambda$ and $R > 0$, define

$$\Theta_R(\Lambda^*) = \sum_{\xi \in \Lambda^* \setminus \{0\}} \prod_{i=1}^{n} \operatorname{sinc}^2(R\xi_i). \tag{12}$$

Here $\operatorname{sinc}(t) = \sin(\pi t)/(\pi t)$, with $\operatorname{sinc}(0) = 1$. By the following lemma, $\Theta_R(\Lambda^*)$ is the relative variance of the translated box count used in Section 2.

**Lemma 4.1.** *Let $t$ be uniform modulo $\Lambda$, and put*

$$J_R(t) = \#\big((t + \Lambda) \cap B_R\big), \qquad \mu_R = R^n / \det \Lambda.$$

*Then*

$$\mathbb{E}[J_R] = \mu_R, \qquad \frac{\operatorname{Var} J_R}{\mu_R^2} = \Theta_R(\Lambda^*), \tag{13}$$

*and*

$$1 + \Theta_R(\Lambda^*) = \frac{\det \Lambda}{R^{2n}} \sum_{w \in \Lambda} \prod_{i=1}^{n} (R - |w_i|)_+. \tag{14}$$

*Proof.* The periodization of $\mathbf{1}_{B_R}$ is bounded on a fundamental domain and hence square integrable. Its mean is $R^n / \det \Lambda$, and its Fourier coefficient at $\xi \in \Lambda^*$ has absolute value

$$\frac{R^n}{\det \Lambda} \prod_i |\operatorname{sinc}(R\xi_i)|.$$

Parseval's identity proves (13), including convergence of the nonnegative series.

Alternatively, expand $J_R^2$ before integrating over a fundamental domain. The two lattice points differ by some $w \in \Lambda$, and the intersection of the corresponding boxes has volume $\prod_i (R - |w_i|)_+$. Thus

$$\mathbb{E}[J_R^2] = \frac{1}{\det \Lambda} \sum_{w \in \Lambda} \prod_i (R - |w_i|)_+.$$

Combining the two expressions for the second moment gives (14). □

## 4.2 From a one-dimensional weight to a cube bound

The overlap identity in Lemma 4.1 reduces the variance estimate to a weighted sum over the original lattice. To bound this sum, we normalize the lattice to $L/\lambda$ and use the weight $f_r$ at $r = R/\lambda$, where, for $r > 0$,

$$f_r(x) = (r - |x|)_+, \qquad A(r) = \sum_{k \in \mathbb{Z}} f_r(k). \tag{15}$$

The normalized lattice has separation at least one in the infinity norm, so we seek a bound of $A(r)^n$ on the total product weight of any set with this separation. Its coordinate projections need

not themselves be separated, so we first cover their weights by weighted unit intervals. The next lemma relates the total cover weight to the maximum weight of a separated subset, and the following lemma bounds this maximum for $f_r$. Taking products of the covers gives half-open unit cubes, each containing at most one point of the original set.

**Lemma 4.2.** *Let $x_1 < \cdots < x_m$ be real numbers with nonnegative weights $w_1, \dots, w_m$. There are nonnegative weights on the intervals $[x_j, x_j + 1)$ that cover each point $x_i$ to total weight at least $w_i$. Their total weight equals the maximum weight of a subset whose points are pairwise at distance at least one.*

*Proof.* Let $p(i)$ be the last index with $x_{p(i)} \le x_i - 1$, or zero if no such index exists. Define

$$F_0 = 0, \qquad F_i = \max\{F_{i-1}, w_i + F_{p(i)}\}.$$

The first option omits $x_i$, while the second selects $x_i$ and uses an optimal subset among the first $p(i)$ points. Hence $F_m$ is the maximum weight of a separated subset.

Give $[x_j, x_j + 1)$ weight $d_j = F_j - F_{j-1} \ge 0$. The intervals containing $x_i$ are precisely those with $p(i) < j \le i$, so their total weight is

$$\sum_{j=p(i)+1}^{i} d_j = F_i - F_{p(i)} \ge w_i.$$

The total interval weight is $\sum_j d_j = F_m$. □

We now apply this covering principle to $f_r$ using the following bound.

**Lemma 4.3.** *For every finite set $P \subset \mathbb{R}$ whose distinct points are at distance at least one,*

$$\sum_{x \in P} f_r(x) \le A(r).$$

*Writing $r = m + u$ with $m = \lfloor r \rfloor$ and $0 \le u < 1$, one has*

$$A(r) = r^2 + u(1 - u).$$

*Proof.* Discard points with zero weight. If $k$ points remain, order them increasingly. Pair the smallest with the largest, the second smallest with the second largest, and so on. Their separation gives

$$\sum_{x \in P} |x| \ge \lfloor k^2/4 \rfloor.$$

Consequently their total weight is at most $kr - \lfloor k^2/4 \rfloor$. For even $k$, this expression is the average of its values at the two neighboring odd integers. For $k = 2j+1$, it equals $(2j+1)r - j(j+1)$. Maximizing over $j \ge 0$ gives $A(r)$. Summing the terms at $0, \pm 1, \dots, \pm m$ gives the displayed formula. □

**Lemma 4.4** (Weighted packing for cubes)**.** *If $P \subset \mathbb{R}^n$ is $\ell_\infty$-separated by at least one, then*

$$\sum_{x \in P} \prod_{i=1}^{n} f_r(x_i) \le A(r)^n. \tag{16}$$

*In particular, with $\lambda = \lambda_1^{(\infty)}(L)$,*

$$1 + \Theta_R(L^*) \le (9/8)^n \frac{\det L}{\lambda^n} \qquad \text{for every } R \ge 8\lambda/9. \tag{17}$$

*Proof.* Only finitely many points contribute to the sum. Apply Lemma 4.2 to the distinct values in each coordinate projection, with weight $f_r$. The cover's total weight equals that of the heaviest separated subset of the projection, which is at most $A(r)$ by Lemma 4.3.

Multiply the coordinate covers. For each point, the product of its covering weights dominates $\prod_i f_r(x_i)$. Sum first over points and then over product boxes. Every half-open box has side length one and contains at most one point of $P$, so it is counted at most once. The total product weight is at most $A(r)^n$, proving (16).

To prove (17), set $r = R/\lambda$ and apply (16) to $L/\lambda$. Equation (14) gives

$$1 + \Theta_R(L^*) \le \frac{\det L}{\lambda^n} \left( \frac{A(r)}{r^2} \right)^n .$$

For $8/9 \le r \le 1$, the ratio is $1/r \le 9/8$. For $r = m + u \ge 1$, it is at most

$$1 + \frac{u(1-u)}{(1+u)^2} \le \frac{9}{8},$$

since $8u(1-u) \le (1+u)^2$ is equivalent to $(3u-1)^2 \ge 0$. □

For $L = \mathbb{Z}^n$ and $R = 4/3$, equality holds in (17). This example explains why the factor $9/8$ enters the list size.

### 4.3 Exact moments for finite shifts

A sufficiently aligned rational grid preserves the same moments exactly when shifts are sampled from the finite quotient used by the algorithm.

**Lemma 4.5.** *Suppose $\Gamma \subseteq G = \delta\mathbb{Z}^n$ and $R/\delta \in \mathbb{Z}$. Fix $z \in G$. For uniform $a \in G/\Gamma$, let*

$$J_a(z) = \#((a - z/2 + \Gamma) \cap B_R), \qquad \mu_R = R^n/D.$$

*Then*

$$\mathbb{E}_a[J_a(z)] = \mu_R, \tag{18}$$

$$\mathbb{E}_a[J_a(z)^2] = (1 + \Theta_R(\Gamma^*))\mu_R^2, \tag{19}$$

$$\mathbb{E}_a[J_a(z)J_a(z')] \le (1 + \Theta_R(\Gamma^*))\mu_R^2 \qquad (z, z' \in G). \tag{20}$$

*Proof.* Every interval of length $R$ on the translate $\delta\mathbb{Z} - z_i/2$ contains exactly $R/\delta$ points. Averaging the coset therefore gives

$$\mathbb{E}_a[J_a(z)] = \frac{\delta^n}{D}(R/\delta)^n = \mu_R.$$

In the square of the count, the difference between two midpoints belongs to $\Gamma$. For $w \in \Gamma$, each coordinate intersection contains exactly $(R - |w_i|)_+/\delta$ points. Its endpoints differ by integer multiples of $\delta$, even when $z_i/2 \notin \delta\mathbb{Z}$. Thus

$$\mathbb{E}_a[J_a(z)^2] = \frac{1}{D} \sum_{w \in \Gamma} \prod_i (R - |w_i|)_+.$$

Use (14) for the second identity. The two second moments are the same for all $z$, so Cauchy–Schwarz proves the mixed bound. □

## 5 A random superlattice with small box variance

To use the moment identities from Section 4 in the list recursion, we need a superlattice whose relative box variance is small at every required width. The weighted cube estimate determines the index needed to achieve this. Fix a candidate scale $\ell$ satisfying

$$\ell \le \lambda \le (1+\eta)\ell, \qquad \eta = n^{-3}. \tag{21}$$

Among the polynomially many scales tried by the algorithm, we analyze one satisfying (21).

We use $h \ge 2$ to control the variance tolerance and the failure probability, and will choose $h = O(\log n)$ in Section 9 to accommodate the full recursion. Put

$$C = \frac{9}{8}, \qquad P_0 = C^n \frac{\det L}{\ell^n} 2^{4h}. \tag{22}$$

The first factor reflects the weighted cube estimate, while the additional factor $2^{4h}$ provides a margin for controlling all required widths simultaneously. Choose a prime $p$ with $P_0 \le p \le 4P_0$. If $L = B\mathbb{Z}^n$, choose a uniform nonzero $u \in \mathbb{F}_p^n$ and define

$$\Gamma = L + \mathbb{Z}\frac{Bu}{p}, \qquad D = \det \Gamma = \frac{\det L}{p}. \tag{23}$$

Then

$$\frac{1}{4}\ell^n C^{-n} 2^{-4h} \le D \le \ell^n C^{-n} 2^{-4h}. \tag{24}$$

The index is always $p$, so $D$ is fixed before $u$ is drawn.

### 5.1 Reducing box variance through the dual lattice

The dual of (23) is

$$\Gamma^* = \{B^{-T} z : z \in \mathbb{Z}^n,\ \langle z, u\rangle = 0 \pmod p\}.$$

A vector outside $pL^*$ survives with probability

$$\alpha_p = \frac{p^{n-1}-1}{p^n - 1} < \frac{1}{p},$$

while every vector in $pL^*$ survives. Consequently,

$$\mathbb{E}_\Gamma[\Theta_R(\Gamma^*)] = \alpha_p \Theta_R(L^*) + (1-\alpha_p)\Theta_R(pL^*) \le \frac{\Theta_R(L^*)}{p} + \Theta_R(pL^*). \tag{25}$$

This calculation was used for Gaussian weights in [GFH26a], and here we apply it to products of squared sinc functions.

The first term of (25) is at most $2^{-4h}$ by Lemma 4.4. We bound the second term, the variance associated with the fine lattice $L/p$, by counting fundamental cells near the boundary of a box.

**Lemma 5.1.** *For all sufficiently large $n$, the parameters above satisfy*

$$\Theta_R(pL^*) \le 2^{-4h} \qquad (R \ge \ell).$$

*The threshold is uniform over the input lattice, the correct scale, and $h \ge 2$.*

*Proof.* By (5),

$$\lambda_n^{(2)}(L) \le \sqrt{n}\frac{\det L}{\lambda^{n-1}}.$$

Apply Lemma 3.1 and (7) to $L$, then divide the resulting basis by $p$. A fundamental cell for $L/p$ is contained in $[-F, F]^n$, where, with $K(n) = n^3 A_{\mathrm{red}}(n)$,

$$F \le K(n)\ell C^{-n} 2^{-4h}. \tag{26}$$

We use this cell only for analysis.

Let $d = \det(L/p)$. For every translate of a box of side $R$, the union of the cells anchored at its lattice points is contained in the box expanded by $F$ in each direction. Conversely, it covers the box contracted by $F$ in each direction. Therefore

$$(R - 2F)_+^n \le dJ_R(a) \le (R + 2F)^n.$$

For $2nF/R \le 1$, the relative deviation from the mean $R^n/d$ is at most $4nF/R$. The variance identity in Lemma 4.1 gives

$$\Theta_R(pL^*) \le 16n^2(F/R)^2 \le 16n^2 K(n)^2 C^{-2n} 2^{-8h}.$$

The last expression is at most $2^{-4h}$ for all sufficiently large $n$. □

**Theorem 5.2** (Small box variance on a random superlattice)**.** *Fix $m$ prescribed widths $R \ge \ell$. For all sufficiently large $n$, the construction (23) satisfies*

$$\Theta_R(\Gamma^*) \le \theta := 2^{-h} \quad \textit{at every prescribed width}$$

*except with probability at most $m2^{1-3h}$.*

*Proof.* Equation (25), together with Lemmas 4.4 and 5.1, gives

$$\mathbb{E}_\Gamma[\Theta_R(\Gamma^*)] \le 2^{1-4h}.$$

Apply Markov's inequality at each prescribed width and take a union bound. □

## 5.2 Choosing the initial scale

To initialize the recursion, we will sample directly in a cube much wider than a fundamental cell of $\Gamma$, so we need a basis whose cell radius is at most a polynomial factor times $\ell$. The small variance bound provides such a basis through a lower bound on the shortest dual vector and transference.

Call the event in Theorem 5.2 $\mathcal{G}$. We include $\ell$ among the prescribed widths. On $\mathcal{G}$, there is no nonzero $\xi \in \Gamma^*$ with $\ell\|\xi\|_2 \le 1/4$. Indeed, the elementary inequality $\sin t \ge t - t^3/6$ gives

$$\prod_i \operatorname{sinc}^2(\ell\xi_i) \ge 1 - \frac{\pi^2}{3}\ell^2\|\xi\|_2^2 \ge 1 - \frac{\pi^2}{48} > \frac{1}{2},$$

contradicting $\Theta_\ell(\Gamma^*) \le 2^{-h}$. By (6),

$$\lambda_n^{(2)}(\Gamma) < 4n\ell.$$

After Lemma 3.1, the fundamental cell has coordinate radius at most

$$F_{\max} = 4n^3 A_{\mathrm{red}}(n)\ell. \tag{27}$$

This bound can be checked from the computed basis by summing the infinity norms of its columns.

The root will have radius $R_0$, and the leaves will have the largest radius $R_t$. Fix

$$\gamma = \frac{19}{20}, \qquad R_0 = (1+\eta)\ell, \qquad R_j = R_0\gamma^{-j}. \tag{28}$$

Let $t$ be the least nonnegative integer such that

$$R_t \geq 8n^5 A_{\mathrm{red}}(n)\ell = 2n^2 F_{\max}. \tag{29}$$

Since $A_{\mathrm{red}}$ is a fixed polynomial, $t = O(\log n)$. Both $t$ and the ratios $R_j/\ell$ depend only on $n$. The widths supplied to Theorem 5.2 are $\ell, R_0, \ldots, R_t$, so they are fixed before the random superlattice is drawn.

# 6 Building the lists

Fix $\Gamma \in \mathcal{G}$ and its preprocessed basis. We construct lists satisfying the reference counting bounds in (11) along a binary tree with leaves at radius $R_t$. Direct sampling establishes the bounds at the leaves, while each shortening step preserves the exact first moment and propagates the bound on the second counting moment. Combining these steps gives a list in $(c+\Gamma)\cap Q_{R_j}$ for any prescribed coset. Each subtree uses private samples, private shifts, and private retention coins.

To align the midpoint boxes and the coordinate partitions used by the searches, choose a single grid $G = \delta\mathbb{Z}^n$ satisfying

$$p^{-1}L \subseteq G, \qquad \ell/(2\delta) \in \mathbb{Z}, \qquad R_j/(6\delta) \in \mathbb{Z} \quad (0 \leq j \leq t), \qquad \delta/\ell \leq n^{-10}. \tag{30}$$

A common denominator followed by integer subdivision gives such a grid with polynomial encoding length. The large size of the quotient $G/\Gamma$ does not affect the cost of sampling one of its elements.

## 6.1 The leaf sampler

At a leaf, the box is wide relative to a fundamental cell, allowing direct sampling with exactly constant expected counts.

Let $P = B_\Gamma[0,1)^n$ be the fundamental cell and use the bound $P \subseteq [-F_{\max}, F_{\max}]^n$. Set

$$U = Q_{R_t+F_{\max}}, \qquad V_U = \mathrm{vol}(U).$$

Draw $X$ uniformly from $U \cap G$. For the prescribed coset $c+\Gamma$, compute

$$Y = c + B_\Gamma \left\lfloor B_\Gamma^{-1}(X-c) \right\rfloor. \tag{31}$$

Then $X \in Y + P$. Keep the proposal only when $Y \in Q_{R_t}$.

**Lemma 6.1.** *For any positive integer $N$, let*

$$M = \left\lceil N\frac{V_U}{(2R_t)^n} \right\rceil, \qquad q_0 = \frac{NV_U}{M(2R_t)^n}.$$

*Make $M$ independent proposals as in (31). Independently keep each successful proposal with probability $q_0$. For every fixed coset $c+\Gamma$, the resulting list satisfies*

$$m(x) = A_t := \frac{ND}{(2R_t)^n}, \qquad S(x,x') \leq A_t^2 \quad (x, x' \in (c+\Gamma)\cap Q_{R_t}).$$

*It uses $O(N)$ proposals, each of polynomial cost.*

*Proof.* For every allowed output $y$, its complete cell $y + P$ lies in $U$. That cell contains exactly $D/\delta^n$ grid points, one for each coset of $\Gamma$ in $G$. Hence a proposal equals $y$ with probability $D/V_U$, independently of $y$ and $c$.

The first moment is $Mq_0D/V_U = A_t$. Distinct retained entries come from distinct proposals, so

$$S(x, x') = M(M-1)(q_0D/V_U)^2 = (1 - 1/M)A_t^2.$$

Finally, (29) gives

$$\frac{V_U}{(2R_t)^n} = \left(1 + \frac{F_{\max}}{R_t}\right)^n \le e^{1/(2n)}.$$

Thus $M = O(N)$. □

### 6.2 One shortening step

Suppose a node is to produce a list in $c + \Gamma$ at radius $\gamma R$. Choose a uniform $a \in G/\Gamma$. The two child cosets are

$$c_+ = a + \Gamma, \qquad c_- = a - c + \Gamma. \tag{32}$$

Generate the two child lists at radius $R$ with independent private randomness. For each ordered pair of child entries, keep the difference with probability

$$q\mathbf{1}_{\{x-y\in Q_{\gamma R}\}}\mathbf{1}_{\{(x+y)/2\in B_R\}}, \qquad q = \frac{(2/\gamma)^n}{N}. \tag{33}$$

Each distinct pair of child identifiers uses an independent coin and creates at most one output entry.

Fix $z \in (c+\Gamma)\cap Q_{\gamma R}$. An admissible pair with difference $z$ is uniquely determined by its midpoint. Its midpoint set is

$$(a - z/2 + \Gamma) \cap B_R.$$

Every such midpoint gives two endpoints in the child boxes, so the number of representations is exactly the variable $J_a(z)$ of Lemma 4.5.

The full differencing recursion of [GHH26, Lemma 4.1] applies here with a constant expected count per point. Its error has two sources: variation in the number of admissible midpoints and reuse of an input entry. The first is controlled by the box variance, while the second contributes a term that decreases with the list size.

**Lemma 6.2** (Counting moments after a difference step)**.** *Suppose both child lists satisfy (11) with common expected count $A = A_R$ and error $\varepsilon$, uniformly over their prescribed cosets. Assume they are independent after the cosets in (32) are fixed, and that $q \le 1$. The output satisfies*

$$m_{\text{out}}(z) = A_{\gamma R}$$

*and*

$$S_{\text{out}}(z, z') \le e^{\varepsilon'} A_{\gamma R}^2, \qquad \varepsilon' = 2\varepsilon + \log\left(1 + \Theta_R(\Gamma^*) + \frac{2^{n+1}}{N}\right). \tag{34}$$

*The bounds hold for every fixed output coset and all $z, z' \in (c + \Gamma) \cap Q_{\gamma R}$.*

*Proof.* Fix $a$. The first moment at $z$ is $qA^2J_a(z)$. Averaging $a$ gives

$$qA^2\mu_R = \frac{(2/\gamma)^n}{N}\left(\frac{ND}{(2R)^n}\right)^2\frac{R^n}{D} = \frac{ND}{(2\gamma R)^n} = A_{\gamma R}.$$

For the second factorial moment, classify two distinct output entries by their child identifiers. There are three cases: four distinct child entries, one shared left entry, or one shared right entry.

Four distinct child entries contribute at most

$$e^{2\varepsilon}q^2A^4\mathbb{E}_a[J_a(z)J_a(z')] \le e^{2\varepsilon}q^2A^4(1+\Theta_R)\mu_R^2.$$

For a shared left entry at $x$, the right values are fixed as $x-z$ and $x-z'$. Since the number of possible triples is at most $J_a(z)$, this contribution is at most

$$e^{2\varepsilon}q^2A^3\mathbb{E}_a[J_a(z)] = e^{2\varepsilon}q^2A^3\mu_R.$$

The same bound holds for a shared right entry. This also covers $z=z'$, since different entries may have equal values. Sharing both child identifiers would give the same output entry and is excluded from the factorial moment.

Divide the sum by $A_{\gamma R}^2 = (qA^2\mu_R)^2$. Since

$$A\mu_R = N/2^n,$$

the result is (34). □

### 6.3 The complete tree

Set $A_j = ND/(2R_j)^n$. A call $\mathsf{Build}(j,c)$ produces a list in $(c+\Gamma)\cap Q_{R_j}$. At $j=t$, it invokes the leaf sampler. Otherwise, it uses (32), calls $\mathsf{Build}(j+1,a)$ and $\mathsf{Build}(j+1,a-c)$ independently, and applies (33) at input radius $R_{j+1}$.

Starting with $\varepsilon_t = 0$, Lemma 6.2 gives

$$\varepsilon_0 \le (2^t-1)\left(\theta + \frac{2^{n+1}}{N}\right). \tag{35}$$

The factor $2^t$ counts the effect of the entire tree. It remains polynomial because $t = O(\log n)$.

The first moment stays exact throughout the construction. If a node's coset is uniform in $G/\Gamma$, then (9) gives

$$\mathbb{E}[|\mathcal{X}_j|] = A_j\frac{\delta^n}{D}\#(Q_{R_j}\cap G) = N. \tag{36}$$

A uniform root coset makes every node coset uniform. The lists below two distinct children need only be independent after their prescribed cosets are fixed.

Algorithm 1 describes the generation rule. The implementation places fixed limits on list sizes and search work. These limits are chosen in Section 9. Until then, the counting analysis refers to the same finite construction without those limits.

**Algorithm 1:** Build$(j, c)$

**Input:** Level $j$, coset representative $c \in G$, fixed $\Gamma$, grid and radii.
**Output:** A list in $(c+\Gamma) \cap Q_{R_j}$, or fail.

**if** $j = t$ **then**
  **return** the list of Lemma 6.1
Choose $a$ uniformly in $G/\Gamma$
$\mathcal{X}_+ \leftarrow$ Build$(j+1, a)$ and $\mathcal{X}_- \leftarrow$ Build$(j+1, a-c)$, using independent randomness
**if** *either call failed* **then**
  **return** fail
Initialize an empty output list
Use the ternary search of Section 7 on $\mathcal{X}_+, \mathcal{X}_-$
**foreach** *reported pair of entries* $(x, y)$ **do**
  **if** $x - y \in Q_{R_j}$ *and* $(x+y)/2 \in B_{R_{j+1}}$ **then**
    With probability $q$, append a new entry with value $x - y$
**return** the output list

# 7 Finding pairs for recursive differencing

To implement the difference rule, we first select candidate pairs by their coordinate classes and then test the exact conditions in (33). A prefix table reports these candidates, and the counting identities from Section 6 bound their expected number.

## 7.1 Three classes per coordinate

At input radius $R$, define

$$b_i(x) = \begin{cases} 0, & |x_i| \le R/3, \\ 1, & R/3 < |x_i| \le 2R/3, \\ 2, & 2R/3 < |x_i| \le R. \end{cases} \tag{37}$$

For an eligible pair,

$$|x_i| + |y_i| = \max\{|x_i - y_i|, |x_i + y_i|\} \le R,$$

so $b_i(x) + b_i(y) \le 2$. Thus, for each left entry $x$, we need to report all right entries whose class vectors satisfy $b_i(y) \le 2 - b_i(x)$ in every coordinate. We next give a table that answers these queries and will also be used for the binary classes in the final search.

## 7.2 Prefix tables and search cost

Consider records with class vectors in $\{0, \dots, s-1\}^n$, where $s$ is either two or three. Given upper bounds $u \in \{0, \dots, s-1\}^n$, a query reports all records whose class vector $b$ satisfies $b_i \le u_i$ in every coordinate.

Let $H(b)$ be the histogram of class vectors, with a list of record identifiers at each occupied vector. For every $0 \le k \le n$, store a table

$$T_k(a_1, \dots, a_k, u_{k+1}, \dots, u_n) = \sum_{b_{k+1} \le u_{k+1}, \dots, b_n \le u_n} H(a_1, \dots, a_k, b_{k+1}, \dots, b_n). \tag{38}$$

Each table has $s^n$ entries and is obtained from $H$ by prefix sums in the last $n-k$ coordinates. All tables therefore take $O(n^2 s^n)$ arithmetic operations and $O(n s^n)$ storage, in addition to the records.

To answer a query, start with the empty prefix. At depth $k$, try the at most $s$ allowed values for the next coordinate and consult $T_{k+1}$. Descend only when the count is positive. At depth $n$, report the identifiers stored at that class vector.

Every visited nonempty prefix has a reported record below it. Charging the prefix to the first such record gives at most $n$ charges per record, each with at most $s$ child tests, so a query costs

$$O(ns(1 + \text{number of reported records})).$$

For the internal search, build the ternary tables on the right list and query with $u_i = 2 - b_i(x)$ for each left entry $x$. Test the exact difference and midpoint conditions on every report. Each pair of entry identifiers is reported at most once, so independent retention coins implement the generation rule of Section 6.

**Proposition 7.1.** *Fix $\Gamma \in \mathcal{G}$ and an internal call whose prescribed output coset is uniform in $G/\Gamma$. If the two child lists have the exact first moments in (11), then their expected number of ternary candidate pairs is*

$$\mathbb{E}[P_3] = N^2 \left( \frac{2}{3} + \frac{2\delta}{3R} - \frac{\delta^2}{4R^2} \right)^n. \tag{39}$$

*Including construction of the tables, the expected search work is at most*

$$\mathrm{poly}(n, \mathrm{size}(B))(3^n + N + N^2(2/3)^n e^{O(n\delta/R)}).$$

*Proof.* Write the parent coset as $c + \Gamma$. For independent uniform $c, a \in G/\Gamma$, the map

$$(c, a) \longmapsto (a, a - c)$$

is a bijection of the quotient squared. Hence the two child cosets are jointly uniform. First fix these cosets and multiply the child first moments, using their private independence. Averaging the cosets then shows that every pair in $(Q_R \cap G)^2$ has expected count

$$A_R^2(\delta^n/D)^2.$$

Let $m = R/\delta$, which is divisible by six. The three one-dimensional class sizes are $2m/3+1$, $2m/3$, and $2m/3-1$. Summing the six combinations with class sum at most two gives the fraction

$$\frac{2}{3} + \frac{2}{3m} - \frac{1}{4m^2}.$$

The free grid has $(2R/\delta)^{2n}$ endpoint pairs, which proves (39) after multiplication by the expected count above. The table construction and query bounds, together with $\mathbb{E}[|\mathcal{X}_\pm|] = N$, give the work estimate. □

A uniform common root coset makes the output coset of every internal call marginally uniform, so Proposition 7.1 applies at every level.

## 8 Recovering a shortest vector

The two final lists lie in the same coset $z + \Gamma$ and are independent conditional on $z$ and $\Gamma$. The objective now changes from generating another list to recovering a shortest vector of $L$ from their differences. We combine quotient labels that identify differences in $L$ with a binary condition that uses the separation of $L$ to control the expected number of reports. After defining the condition, we show that it retains enough representations of a shortest vector, bound all reported pairs, and give the search algorithm.

### 8.1 Quotient labels and a binary condition

For $x = z + g_x$ with $g_x \in \Gamma$, give the entry label

$$\mathrm{lab}(x) = g_x + L \in \Gamma/L. \tag{40}$$

Then

$$\mathrm{lab}(x) = \mathrm{lab}(y) \quad \Longleftrightarrow \quad x - y \in L.$$

A label is represented by one field element, as detailed in Appendix B.

Let

$$I = [-R_0, R_0), \qquad B_0 = [-\ell/2, \ell/2).$$

For each endpoint, store the binary class vector

$$h_i(x) = \mathbf{1}_{\{x_i \notin B_0\}}.$$

The final search reports exactly the pairs of entries satisfying

$$\mathrm{lab}(x) = \mathrm{lab}(y), \qquad h_i(x) + h_i(y) \le 1 \quad (1 \le i \le n). \tag{41}$$

Thus every coordinate has at least one endpoint in the inner interval $B_0$. Each reported nonzero difference is tested for membership in $L$ and compared with the current best vector.

### 8.2 A shortest vector has many compatible representations

Fix a shortest vector $v$. The full midpoint box $B_\ell$ gives endpoints

$$x = M + v/2, \qquad y = M - v/2$$

in $Q_{R_0}$. Let $J_{\mathrm{full}}(z)$ count its midpoints in $z - v/2 + \Gamma$, and let $J(z)$ count those also satisfying the binary condition in (41).

**Lemma 8.1.** *For every $\Gamma \in \mathcal{G}$, a uniform $z \in G/\Gamma$ satisfies*

$$J(z) \ge \frac{1}{4}\frac{\ell^n}{D} \tag{42}$$

*except with probability at most $4\theta + 4n\eta$.*

*Proof.* Put $\mu = \ell^n/D$. By Lemma 4.5,

$$\mathbb{E}_z[J_{\mathrm{full}}] = \mu, \qquad \mathrm{Var}_z\, J_{\mathrm{full}} \le \theta\mu^2.$$

Chebyshev's inequality gives $J_{\mathrm{full}} \ge \mu/2$ except with probability $4\theta$.

Consider one coordinate. If $|v_i| \le \ell$, every midpoint in $[-\ell/2, \ell/2)$ puts at least one endpoint in $B_0$. If $|v_i| > \ell$, the excluded midpoints form a central interval of length $|v_i| - \ell \le \eta\ell$. The endpoints of this interval align with the midpoint grid because $v_i \in \delta\mathbb{Z}$ and $\ell/(2\delta) \in \mathbb{Z}$.

A union bound over coordinates gives an expected deletion count of at most $n\eta\mu$, so Markov's inequality bounds the probability of deleting more than $\mu/4$ by $4n\eta$. On the complementary events, (42) holds. □

To turn these midpoint locations into actual pairs, fix a root coset with $J \geq 1$ and let $Z$ count pairs of entries with difference $v$ and a compatible midpoint. The two root lists are independent with the moments in (11), so $\mathbb{E}[Z] = A_0^2 J =: M$. Since each endpoint determines the other uniquely, two different midpoint locations have different endpoint values on both sides and contribute at most $e^{2\varepsilon_0} A_0^4$ to the second moment. At a repeated location, the ordinary second moment on each side is at most $e^{\varepsilon_0}(A_0^2 + A_0)$. Consequently,

$$\mathbb{E}[Z^2] \leq e^{2\varepsilon_0}(A_0^4 J^2 + 2A_0^3 J + A_0^2 J),$$

and the second-moment inequality gives

$$\mathbb{P}[Z > 0] \geq \frac{e^{-2\varepsilon_0}}{1 + 2/(A_0 J) + 1/M} \geq \frac{e^{-2\varepsilon_0}}{(1 + M^{-1/2})^2}, \tag{43}$$

where the last inequality uses $A_0 J = \sqrt{MJ} \geq \sqrt{M}$.

On (42), the expected number of these pairs satisfies

$$A_0^2 J \geq \frac{N^2 D \ell^n}{4(2R_0)^{2n}}. \tag{44}$$

The polynomial factors selected in Section 9 make this quantity tend to infinity.

### 8.3 Bounding the number of reported pairs

We now bound all pairs satisfying (41), including zero differences. For the geometric bound, we will express the sum over differences in $L$ using nonnegative weights on translates of $B_\ell$. Each translated cube contains at most one point of $L$. We first derive the one-dimensional identity behind this expression.

Write

$$I_\delta = I \cap \delta\mathbb{Z}, \qquad B_\delta = B_0 \cap \delta\mathbb{Z}.$$

For $w \in \delta\mathbb{Z}$, let $c_\delta(w)$ be $\delta$ times the number of compatible one-dimensional endpoint pairs at displacement $w$:

$$c_\delta(w) = \delta \sum_{x \in \delta\mathbb{Z}} \mathbf{1}_{I_\delta}(x)\mathbf{1}_{I_\delta}(x - w)\mathbf{1}_{\{x \in B_\delta \text{ or } x - w \in B_\delta\}}. \tag{45}$$

Inclusion and exclusion counts pairs with the first endpoint in $B_\delta$, adds those with the second endpoint there, and subtracts those with both endpoints there. The first two counts agree because $I_\delta$ and $B_\delta$ are symmetric about the same point $-\delta/2$. Thus, with

$$g = 2\mathbf{1}_{I_\delta} - \mathbf{1}_{B_\delta} \geq 0,$$

we obtain

$$c_\delta(w) = \delta \sum_{x \in \delta\mathbb{Z}} \mathbf{1}_{B_\delta}(x) g(x - w). \tag{46}$$

The weight $g$ is one on $B_\delta$ and two on $I_\delta \setminus B_\delta$, so its total mass is

$$\delta \sum_{x \in \delta\mathbb{Z}} g(x) = 4R_0 - \ell.$$

Taking products over coordinates gives the factor in the following bound.

**Lemma 8.2** (Final report bound)**.** *If $\ell \le \lambda_1^{(\infty)}(L)$, then*

$$\sum_{w\in L}\prod_{i=1}^{n} c_\delta(w_i) \le (4R_0-\ell)^n. \tag{47}$$

*For the two root lists generated in a uniform common coset of a fixed $\Gamma \in \mathcal{G}$, the number $P$ of pairs satisfying (41) obeys*

$$\mathbb{E}[P] \le \frac{A_0^2}{D}(4R_0-\ell)^n. \tag{48}$$

*Proof.* Apply (46) in every coordinate, then set $t = x - w$. All summands are nonnegative, so

$$\sum_{w\in L}\prod_{i} c_\delta(w_i) = \delta^n \sum_{t\in G}\prod_{i} g(t_i)\sum_{w\in L}\mathbf{1}_{B_\delta^n}(t+w).$$

The inner sum is at most one. Two points in the half-open cube $B_\delta^n$ differ by less than $\ell$ in every coordinate, while a nonzero vector of $L$ has infinity norm at least $\ell$. Therefore

$$\sum_{w\in L}\prod_{i} c_\delta(w_i) \le \delta^n \sum_{t\in G}\prod_{i} g(t_i) = \left(\delta\sum_{x\in\delta\mathbb{Z}} g(x)\right)^n = (4R_0-\ell)^n.$$

For the work estimate, first fix the common coset $z+\Gamma$. The two root lists use independent private randomness, and their first moments are exactly $A_0$. If $x - y = w \in L$, then membership of $x$ in $z+\Gamma$ already implies membership of $y$. Averaging $z$ therefore contributes one factor $\delta^n/D$.

For a fixed $w$, the number of compatible free grid pairs is $\delta^{-n}\prod_i c_\delta(w_i)$. Thus

$$\mathbb{E}[P] = \frac{A_0^2}{D}\sum_{w\in L}\prod_{i} c_\delta(w_i),$$

which proves (48). The term $w = 0$ is included throughout. □

To see the $3^n$ balance already at this point, put $r = R_0/\ell$. Our parameters give $N^2D/\ell^n = 4^n\,\mathrm{poly}(n)$, so the report bound is

$$\mathbb{E}[P] \le \mathrm{poly}(n)\left(\frac{4r-1}{r^2}\right)^n \le 3^n\,\mathrm{poly}(n), \qquad r \ge 1.$$

The last inequality is $4r - 1 \le 3r^2$. The remaining task is to find these pairs without scanning all pairs in a large label group.

## 8.4 Searching within label groups

To turn the report bound into a running time bound, we must also control the cost of searching within each label group. A split by group size balances direct comparisons against the cost of building a binary table.

**Proposition 8.3.** *Let two fixed labeled lists contain $M$ entries in total, and let $P$ be the number of their pairs satisfying (41). There is a deterministic algorithm that reports every such pair of entries once, in time*

$$\mathrm{poly}(n, \log(M+2), \textit{label length})(M2^{n/2} + P) \tag{49}$$

*and space* $\mathrm{poly}(n)(M + 2^n)$.

*Proof.* Sort the entries by label. Put $H = \lceil 2^{n/2} \rceil$, and let $m_\xi$ be the total size of the two sides in label group $\xi$.

If $m_\xi \le H$, check all cross pairs in that group. The total cost of these groups is at most $\sum_\xi H m_\xi \le HM$.

There are at most $M/H$ groups with $m_\xi > H$. In each such group, build the binary prefix tables of Section 7.2 on the right side and query with bounds $u_i = 1 - h_i(x)$ for each left entry $x$. The total table and query cost is

$$\mathrm{poly}(n) \left( \frac{M}{H} 2^n + M + P \right).$$

Together with the small groups and sorting, this gives (49). Process the large groups one at a time to obtain the space bound. □

# 9 The algorithm and its analysis

We now choose the polynomial factors, remove knowledge of $\lambda$, and place finite limits on every trial. The list construction and the two search bounds then give the claimed complexity.

## 9.1 Parameters and the scale grid

The constants and radii are those of (28) and (29). We choose the variance tolerance and the polynomial factor in the list size together to keep the accumulated moment error small and make the expected number of target pairs large. Let

$$Q = 2^{t+2}, \qquad h = \left\lceil \log_2 (Q(n+2)^{10}) \right\rceil, \qquad N = 2^{3h} \left\lceil \sqrt{(9/2)^n} \right\rceil. \tag{50}$$

Here $Q$ exceeds the total number of nodes in the two trees. Since $t = O(\log n)$,

$$Q = \mathrm{poly}(n), \qquad h = O(\log n), \qquad (3/\sqrt{2})^n 2^{3h} \le N \le 2(3/\sqrt{2})^n 2^{3h}.$$

Since $\gamma\sqrt{C} > 1$, the retention probability in (33) satisfies

$$q \le 2^{-3h} (\gamma\sqrt{C})^{-n} < 1.$$

On every good superlattice, (35) gives

$$\varepsilon_0 \le Q \left( 2^{-h} + \frac{2^{n+1}}{N} \right) \le (n+2)^{-10} + 2Q^{-2}(n+2)^{-30}. \tag{51}$$

Set

$$\ell_{\mathrm{lo}} = \|B^{-1}\|_{\infty\to\infty}^{-1}, \qquad \ell_{\mathrm{hi}} = \min_i \|b_i\|_\infty.$$

Then $\ell_{\mathrm{lo}} \le \lambda \le \ell_{\mathrm{hi}}$. We try every scale $\ell_j = \ell_{\mathrm{lo}}(1+\eta)^j \le \ell_{\mathrm{hi}}$. This gives

$$O(1 + n^3 \log(\ell_{\mathrm{hi}}/\ell_{\mathrm{lo}})) = \mathrm{poly}(n, \mathrm{size}(B))$$

scales of polynomial encoding length, at least one of which satisfies (21).

At each scale, set $Y = \max\{2, \lceil P_0 \rceil\}$ and test $O((n+h+1)\log(2Y))$ independent uniform candidates in $[Y, 2Y]$ by deterministic primality testing [AKS04]. Standard prime density bounds make the failure probability at most $2^{-n-10}$, and every prime returned on a correct scale satisfies $P_0 \le p \le 4P_0$.

### 9.2 A trial with resource limits

Choose the list limit

$$L_{\max} = N2^{2h}. \tag{52}$$

Let $p_{\mathrm{op}}(n, \mathrm{size}(B))$ bound the polynomial overhead of the arithmetic and search operations, and set the work limit for each pair search to

$$W_{\max} = 3^n 2^{8h} p_{\mathrm{op}}(n, \mathrm{size}(B)). \tag{53}$$

Stop the current trial before exceeding either limit, including the cost of table construction and pair reporting. Retain every lattice vector already found as a candidate.

**Algorithm 2: Shortest vector search in the infinity norm**

**Input:** A nonsingular rational matrix $B$, with $L = B\mathbb{Z}^n$.
**Output:** A nonzero lattice vector $v_{\mathrm{best}}$.

Initialize $v_{\mathrm{best}}$ with a shortest column of $B$ in the infinity norm
Compute $t, Q, h, N$ from (29) and (50)
**foreach** *scale $\ell$ in the multiplicative grid* **do**
  Compute $R_0, \dots, R_t$, $F_{\max}$, and $P_0$
  Run the bounded prime search. Skip this scale if it fails
  Choose uniform nonzero $u \in \mathbb{F}_p^n$ and form $\Gamma = L + \mathbb{Z}Bu/p$
  Apply Lemma 3.1 to $\Gamma$
  **if** *the sum of the infinity norms of the basis columns exceeds $F_{\max}$* **then**
    Skip this scale
  Choose $G = \delta\mathbb{Z}^n$ satisfying (30), and choose uniform $z \in G/\Gamma$
  Run two independent calls $\mathcal{X}_+ \leftarrow \mathsf{Build}(0, z)$ and $\mathcal{X}_- \leftarrow \mathsf{Build}(0, z)$ with limits (52) and (53)
  **if** *either call fails* **then**
    Skip this scale
  Assign quotient labels and run the search of Proposition 8.3, with work limit $W_{\max}$
  **foreach** *reported pair $(x, y)$* **do**
    Set $w = x - y$
    **if** *$w \neq 0$, $B^{-1}w \in \mathbb{Z}^n$, and $\|w\|_\infty < \|v_{\mathrm{best}}\|_\infty$* **then**
      Set $v_{\mathrm{best}} \leftarrow w$
**return** $v_{\mathrm{best}}$

We analyze Algorithm 2 with exact random choices. Appendix B implements these choices within fixed resource bounds, with total additional failure probability at most $2^{-n-10}$.

### 9.3 Success of the unrestricted construction

Fix a shortest vector $v$ and a correct scale. Conditional on a successful prime search, Theorem 5.2 gives

$$\mathbb{P}[\mathcal{G}^c] \le (t+2)2^{1-3h} = o(1).$$

On $\mathcal{G}$, the basis check passes. By Lemma 8.1, the root shift has $J \ge \ell^n/(4D)$ except with probability $4\theta + 4n\eta = o(1)$.

For every such root shift, combine (24), (44), and (50):

$$A_0^2 J \ge \frac{2^{2h}}{16(1+\eta)^{2n}} =: M_*. \tag{54}$$

Since $M_* \to \infty$, the moment bounds in (11) and (51), together with independence conditional on the common root coset, allow us to apply (43). The two trees therefore contain a pair with difference $v$ satisfying the binary condition with probability at least

$$\frac{e^{-2\varepsilon_0}}{(1+M_*^{-1/2})^2} = 1 - o(1).$$

The deterministic final search reports every such pair unless its resource limit is reached.

### 9.4 Bounding the probability of interruption

All expectations in this subsection fix a good $\Gamma$ and average the root shift and the private tree randomness. They refer to the construction without resource limits.

Every node has expected list size $N$ by (36). A union bound and Markov's inequality give

$$\mathbb{P}[\text{some list exceeds } L_{\max}] \le Q 2^{-2h} = o(1). \tag{55}$$

After absorbing constants into $p_{\mathrm{op}}$, Proposition 7.1 and $\delta/R \le n^{-10}$ bound the expected work of each internal search by

$$p_{\mathrm{op}}(n, \mathrm{size}(B))\, 3^n 2^{6h}.$$

For the final search, (48) and the parameter choices give

$$\mathbb{E}[P] \le 4 \cdot 2^{2h} \left(\frac{3+4\eta}{(1+\eta)^2}\right)^n \le 4 \cdot 3^n 2^{2h}.$$

Also,

$$\mathbb{E}[M]\, 2^{n/2} = 2N 2^{n/2} \le 4 \cdot 3^n 2^{3h},$$

so Proposition 8.3 gives the same expected work bound for the final search. The work limit (53) leaves a factor of $2^{2h}$, and Markov's inequality followed by a union bound over the searches gives

$$\mathbb{P}[\text{some search reaches its work limit before a list limit}] \le O(Q 2^{-2h}) = o(1). \tag{56}$$

These bounds do not condition on the presence of the target pair.

To compare the implementation with the unrestricted construction, give both the same random choices until the first limit is reached. They produce identical records before that point. Equations (55) and (56) bound the probability of this first interruption. Combined with the preceding correctness argument, one correct scale trial succeeds with probability $1 - o(1)$.

### 9.5 Total time, space, and amplification

There are polynomially many scales and at most $Q = \mathrm{poly}(n)$ tree nodes per scale. Since $h = O(\log n)$, leaf sampling and pair searches take $\widetilde{O}(3^n)$ time under the stated resource limits, as summarized in Table 1. The $2^{n+o(n)}\,\mathrm{poly}(\mathrm{size}(B))$ basis preprocessing cost per scale and the polynomial overhead of the remaining operations are absorbed in this bound.

Evaluating the trees in depth-first order keeps at most $O(t)$ completed lists waiting, so the live lists use $O(tN2^{2h})$ records. An internal search uses $O(n3^n)$ table entries, while the final search needs only $O(n2^n)$ additional table entries because large label groups are processed one at a time. Since each record and each table count has polynomial encoding length, the total space is $\widetilde{O}(3^n)$.

*Proof of Theorem 1.1.* On a correct scale, the preceding bounds give total failure probability $o(1)$, including the finite randomness loss. Thus a basic execution succeeds with probability at least $2/3$ for all sufficiently large $n$.

When the correct scale succeeds, its final search reports a shortest vector. Every update of $v_{\mathrm{best}}$ is an exactly checked nonzero vector of $L$, so a later scale cannot replace it by an invalid shorter vector. This proves the success probability of a basic execution. The finitely many smaller dimensions can use the $\ell_\infty$-SVP algorithm of [DPV11], with their costs included in the absolute constant.

Independent repetition $O(n \log n)$ times, returning the shortest output, reduces the failure probability to $2^{-\Omega(n \log n)}$ and preserves the $\widetilde{O}(3^n)$ time and space bounds. □

## 10 Conclusion and discussion

We gave a classical randomized algorithm for exact $\ell_\infty$-SVP in $\widetilde{O}(3^n)$ time and space. Restricting midpoints to a fixed cube preserves equal expected counts through repeated differencing, while second counting moments control the dependence within the lists. Two coordinate searches generate the lists and recover a shortest vector within these bounds.

The lattice kissing number gives a geometric interpretation of the factor $3^n$. In the infinity norm, a lattice has at most $3^n - 1$ shortest nonzero vectors [Henk02, Equation (1.3)], and $\mathbb{Z}^n$ attains this bound with the vectors in $\{-1, 0, 1\}^n \setminus \{0\}$. Including the origin, $L \cap [-\lambda, \lambda]^n$ therefore contains at most $3^n$ points, the same exponential scale as our running time bound. This connection becomes more concrete in the terminal search, where each lattice difference is weighted according to its compatible representations. It is this weighted count that controls the number of pair reports, and the integer lattice again attains the bound. When $L = \mathbb{Z}^n$ and $\ell = 1$, every translate of the half-open cube $[-1/2, 1/2)^n$ contains exactly one lattice point. Consequently, the counting inequality in the proof of Lemma 8.2 becomes an equality, giving on an aligned grid

$$\sum_{w \in \mathbb{Z}^n} \prod_{i=1}^{n} c_\delta(w_i) = (4R_0 - 1)^n.$$

This equals $3^n$ at $R_0 = 1$, and $3^n(1 + o(1))$ for the algorithm's choice $R_0 = 1 + n^{-3}$.

Our analysis guarantees enough representations of a fixed shortest vector, while the terminal search reports all compatible pairs. For the extremal lattice $\mathbb{Z}^n$, however, the large report count comes with many shortest vectors, any one of which would solve SVP. The same abundance that makes complete reporting expensive may therefore help a search whose only goal is to find one shortest vector. This leads to a natural question: can the abundance of shortest vectors be exploited to find one without reporting all compatible pairs?

## A Basis preprocessing

For completeness, we give the specialization of [GHH26, Appendix A.3] that proves Lemma 3.1, using the DBKZ theorem of [ALNS20, Theorem 2.3] and the exact Euclidean SVP algorithm of [MV13].

*Proof of Lemma 3.1.* Suppose $n \geq 4$ and put $k = \lfloor n/2 \rfloor$. Fix a prefix $F = (b_1, \ldots, b_i)$ of a basis $(F, D)$ of $\Lambda$. Its orthogonal projection and remaining lattice are

$$\pi = I - F(F^{\mathsf{T}}F)^{-1}F^{\mathsf{T}}, \qquad \Lambda_i = (\pi D)\mathbb{Z}^d = \pi\Lambda, \qquad d = n - i,$$

with $\pi = I$ when $i = 0$. Clear one common positive denominator in $\pi D$ to obtain the integer input required by [ALNS20]. This preserves length ratios and coefficients, so an output with coefficient vector $z$ lifts to $Dz$.

If $d > k$, apply [ALNS20, Theorem 2.3] with block dimension $k$, exact oracle factor $\delta = 1$, and slack $\varepsilon = 1$. It returns a nonzero projected vector $v$ with

$$\|v\|_2 \le 2\gamma_k^{(d-1)/(2(k-1))} \det(\Lambda_i)^{1/d} \le 2k^{(d-1)/(2(k-1))} \det(\Lambda_i)^{1/d},$$

using polynomially many exact SVP calls of rank $k$. Here $\gamma_k \le k$ is the Euclidean Hermite bound, and the determinant is taken within the span of $\Lambda_i$. If $d \le k$, use exact SVP on $\Lambda_i$.

Divide the output coefficient vector by the gcd of its coordinates to make it primitive, then use the extended Euclidean algorithm to obtain a unimodular completion. Lift this completion and size reduce the selected column against the fixed prefix to obtain $b_{i+1}$, with $\|b_{i+1}^*\|_2 \le \|v\|_2$. The projections of $n$ independent vectors of length at most $\lambda_n^{(2)}(\Lambda)$ span the remaining space. Selecting $d$ independent projections and applying Hadamard's inequality gives

$$\det(\Lambda_i)^{1/d} \le \lambda_d^{(2)}(\Lambda_i) \le \lambda_n^{(2)}(\Lambda).$$

Later operations keep the prefix fixed, so its Gram–Schmidt vectors are preserved. Writing $n = 2k+r$, $r \in \{0, 1\}$, and using $k^{1/(k-1)} \le 2$ for $k \ge 2$, we obtain

$$\max_i \|b_i^*\|_2 \le 2k^{(n-1)/(2(k-1))} \lambda_n^{(2)}(\Lambda) = 2k\, k^{(r+1)/(2(k-1))} \lambda_n^{(2)}(\Lambda) \le 2n\lambda_n^{(2)}(\Lambda).$$

For $n \le 3$, exact SVP at every suffix gives the stronger factor one. Thus $A_{\mathrm{red}}(n) = 2(n+1)$ works in every dimension.

For the encoding bounds, begin with an LLL-reduced input basis. After each selection, LLL reduce the projected suffix, lift its integer column operations, and size reduce against earlier columns, as in [MV13, Section 2]. An LLL-reduced suffix has Gram–Schmidt lengths at most $2^{n/2}\lambda_n^{(2)}(\Lambda)$. Consequently every current basis column has length at most $\sqrt{1+n/4}\max\{2n, 2^{n/2}\}\lambda_n^{(2)}(\Lambda)$. Its denominator divides the original common denominator, and $\lambda_n^{(2)}(\Lambda)$ is at most the longest input column. These bounds give polynomial encoding length at the start of each round. Gram inverse and cofactor formulas give the same bound for projections and integer coordinates. Each reduction, primitive completion, and lift has polynomial intermediate size.

There are $n$ rounds and polynomially many exact SVP calls of rank at most $k$. The algorithm of [MV13] uses $4^{k+o(k)}$ time and $2^{k+o(k)}$ space per call, with polynomial input overhead. The overall time and workspace are therefore bounded by $2^{n+o(n)}\,\mathrm{poly}(\mathrm{size}(B))$, including the fixed small dimensions. □

# B Finite arithmetic and random choices

We record the finite representations used by the algorithm and the coupling that makes every execution obey a deterministic resource bound.

## B.1 Constructing the superlattice and its quotient

Lift $u \in \mathbb{F}_p^n$ to an integer vector. A Hermite basis of

$$K = p\mathbb{Z}^n + \mathbb{Z}u$$

is obtained from $[pI \mid u]$ and gives a rational basis of $\Gamma$ after multiplication by $B/p$.

For a terminal entry $x = z + g$, let

$$a_x = pB^{-1}g \in \mathbb{Z}^n.$$

Modulo $p$, this vector lies in the span of $u$. Choose an index $j$ with $u_j \neq 0$ and represent the label by

$$\mathrm{lab}(x) = (a_x)_j u_j^{-1} \pmod{p}. \tag{57}$$

This scalar represents the class $g + L$, so equal labels are equivalent to $x - y \in L$ and can be computed by exact rational and finite field arithmetic.

### B.2 Choosing the grid and sampling its quotient

Choose the mesh in (30) from a common denominator for $B/p$, $\ell/2$, and $R_j/6$, then subdivide until $\delta/\ell \leq n^{-10}$. Its encoding length is polynomial because there are $O(\log n)$ radii of polynomial encoding length. The inclusion $p^{-1}L \subseteq G$ makes this grid valid for every possible $\Gamma$ at the given scale and prime.

Let $H$ be an upper triangular integer Hermite basis of $\Gamma/\delta$. If its positive diagonal entries are $h_1, \ldots, h_n$, the integer box

$$\{0, \ldots, h_1 - 1\} \times \cdots \times \{0, \ldots, h_n - 1\}$$

is the standard set of representatives of $\mathbb{Z}^n/H\mathbb{Z}^n$. Independent uniform coordinates in these intervals, multiplied by $\delta$, therefore sample $G/\Gamma$ in polynomial time.

All node offsets can be reduced to these representatives after each subtraction without changing their cosets or the midpoint identities.

### B.3 Bounded randomness and bit complexity

All random choices reduce to rejection sampling of uniform integers, with acceptance probability at least $1/2$ per attempt. The resource limits give a deterministic bound $R_{\mathrm{rand}}$ on the number of calls, with $\log R_{\mathrm{rand}}$ polynomial in the input length. Truncate each call after

$$s = \lceil \log_2 R_{\mathrm{rand}} \rceil + n + 20$$

attempts and fail the trial if none succeeds. Coupling to the exact choices gives a total disagreement probability at most

$$R_{\mathrm{rand}} 2^{-s} \leq 2^{-n-20}.$$

This includes quotient and field samples, prime candidates, and rational retention coins.

Each record stores a vector, a quotient label, and an identifier local to its node. Vector coordinates lie in $\delta\mathbb{Z}$ and have absolute value at most $R_t$. The grid, radii, quotient labels, identifiers, and table counts all have polynomial encoding length under the parameter and resource bounds. Exact membership tests use $B^{-1}w \in \mathbb{Z}^n$, and all arithmetic and comparisons have polynomial bit cost, as included in $p_{\mathrm{op}}$.

## AI methodology

Building on Gaussian difference dynamics (GDD) [GHH26], the author used GPT-6-astra to assist with extending the framework to the infinity norm, including the algorithm design and mathematical analysis in Sections 4 to 9. The author then used GPT-6.1-sol and GPT-6-astra to help draft the manuscript. The author thoroughly checked and revised the entire paper and remains responsible for all mathematical claims, proofs, and references.

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